Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Project Overview

This analysis examines customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior.

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Dataset Summary

3,900

Purchase Records

Complete transactional data analyzed

18

Data Columns

Customer demographics, purchase details, and shopping behavior

37

Missing Values

Review ratings imputed using median by category

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Data Preparation Process

01

Data Loading & Exploration

Imported dataset using pandas, checked structure with df.info() and summary statistics with .describe()

02

Missing Data Handling

Imputed 37 missing review ratings using median rating by product category

03

Column Standardization

Rewrote columns to snake case for better readability and documentation

04

Feature Engineering

Created age_group by binning ages and purchase_frequency_days from purchase data

05

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis

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Key SQL Analysis Findings

Revenue by Gender

Male customers generated $157,890 vs. Female at $75,191

Shipping Comparison

Express shipping: $60.48 avg vs. Standard: $58.46 avg

Top Products by Rating

Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78)

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Subscription & Customer Segmentation

Subscription Status

- 1,053 subscribers (27%)
- 2,847 non-subscribers (73%)
- Average spend: $59.49 (Yes) vs. $59.87 (No)
- Total revenue: $62,645 (Yes) vs. $170,436 (No)

Customer Segments

- 3,116 Loyal customers
- 701 Returning customers
- 83 New customers

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Product Performance Insights

Highest Discount Rate

50% of hat purchases used discounts

Second Highest

Sneakers at 49.66% discount rate

Third Highest

Coat at 49.07% discount rate

839 customers used discounts but spent above average purchase amount ($59.76)

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Revenue by Age Group

26%

Young Adult

$62,143 total revenue

25%

Middle-aged

$59,197 total revenue

24%

Adult

$55,978 total revenue

24%

Senior

$55,763 total revenue

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Power BI Dashboard

Key KPIs

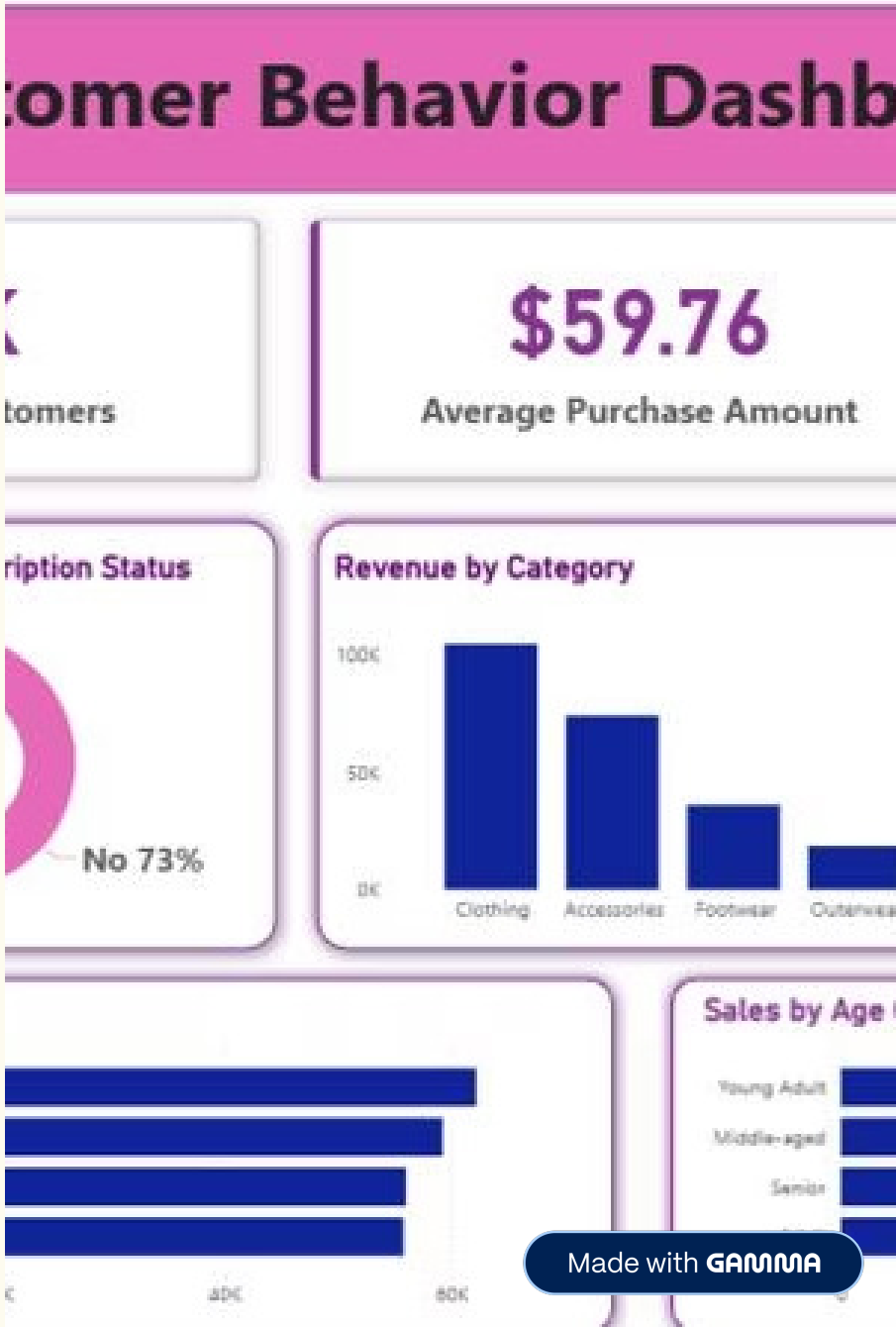
3,900 customers, $59.76 average purchase, 3.75 average rating

Interactive Filters

Subscription status, gender, category, shipping type

Visualizations

Donut charts, bar charts showing revenue and sales by category and age group



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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits to increase subscriber base

Customer Loyalty Programs

Reward repeat buyers to move them into loyal segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products in campaigns

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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